

Two friends, Aiza and Bushra, want to meet in a *fully-observable* city that can be represented as an $M \times N$ grid. They can start in any two squares of the grid. In each time step, they can move UP, DOWN, LEFT, and RIGHT, or they can choose to STAY in the same square. The actions of Aiza and Bushra occur *simultaneously*. They want to devise the most efficient path to meet each other in the shortest possible time. Note that they cannot leave the grid and simply passing each other does not count. To meet, they must be in the same square at the same time.

1. Formulate the problem as a single-agent search by identifying the state space, actions, successor function, and goal test. 8
(CO2)
(PO2)

Solution:

State space: Each state will consist of the coordinates of Aiza and Bushra:

$$((X_A, Y_A), (X_B, Y_B))$$

Actions: Up, Down, Left, Right, and Stay for each of them

Successor Function: Take the current state and action pairs, then possibly change the coordinates of each person.

Goal Test: $(X_A, Y_A) = (X_B, Y_B)$?

Rubric:

- 2 points for each item

2. Among the following graph search algorithms, identify the ones that are guaranteed to find the most efficient path: 3
(CO1)
(PO1)

- Algorithm 1: Depth-First Search
- Algorithm 2: Breadth-First Search
- Algorithm 3: Uniform Cost Search
- Algorithm 4: A* Search (with an admissible but not consistent heuristic)
- Algorithm 5: A* Search (with a consistent heuristic)
- Algorithm 6: A* Search (with a heuristic that returns zero for each state)

Solution:

Algorithms 2, 3, 5, 6

Rubric:

- 0.5 points for each correct identification
- 0.5 points for each correct omission

3. Did you choose to include the STAY action in your search problem formulation? Justify your choice. 2
(CO2)
(PO2)

Solution:

If STAY action was excluded, there may not be solutions to some grids. For example, in a 1×2 grid, where Aiza and Bushra start at different locations, they would swap places forever.

4. Propose a non-trivial admissible heuristic for the problem.

2
(CO3)
(PO3)

Solution:

If one person could move, Manhattan Distance would be a possible choice. However, since both persons can move, it will be inadmissible for certain cases. For example,

A	
	B

Here, the heuristic function will return 2, but the actual cost is 1. Hence, $\frac{\text{Manhattan Distance}}{2}$ could be a possible choice.

Rubric:

- 1 point for non-trivial heuristic
- 1 point for admissibility